PHY-201 Final Examination (Revised Draft) - Fall 2026

SECTION A: MECHANICS

1. A flywheel of moment of inertia 4 kg m^2 is brought to rest from 30 rad/s by a constant torque in 10 s. Determine the torque and the work done. (10 marks)

2. Analyse the motion of a cylinder rolling without slipping down an incline and compare its speed with that of a sliding block. (10 marks)

3. A satellite orbits the Earth in a circular path. Justify, using energy arguments, why it does not need fuel to stay in orbit. (10 marks)

SECTION B: THERMODYNAMICS

4. Design an experiment to measure the specific heat of an unknown metal and evaluate its main sources of error. (10 marks)

5. Compare the efficiencies of a real heat engine and an ideal Carnot engine operating between 500 K and 300 K. (10 marks)